Kane Pavlovich

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Professional Profile

PhD Research Scientist in Computational Neuroscience with 4+ years of experience applying advanced machine learning and statistical modeling to large-scale, high-dimensional data. Proven record of improving predictive accuracy, streamlining data pipelines, and delivering actionable insights across interdisciplinary teams. Strong communicator with experience influencing research directions, mentoring teams, and presenting findings to scientific and non-technical audiences.

Skills

- **Programming:** Python (Pandas, NumPy, Scikit-learn), R (tidyverse, lavaan), MATLAB, SQL, Linux/Bash, Git
- **Machine Learning:** Predictive Modelling (Regression, Classification), Feature Engineering, Kernel Methods, SVM, Ensemble Methods, Dimensionality Reduction (PCA), Time-Series Analysis, Cluster Analysis
- **Statistical Analysis:** Psychometrics, Structural Equation Modelling (SEM), Hypothesis Testing, Signal Processing (Fourier & Wavelet Analysis)
- **Tools & Platforms:** Tableau, Slurm (HPC), GitHub, Jupyter Notebooks
- **Leadership:** Cross-Functional Project Leadership, Mentorship, Curriculum Development, Public Speaking

Experience

PHD RESEARCHER | MONASH UNIVERSITY | JANUARY 2022 - JULY 2025

- **Engineered and validated predictive models** (Kernel Ridge Regression, SVMs) to quantify brain-behavior relationships, **improving prediction accuracy by 15%** over previous benchmarks through novel feature engineering.
- **Developed scalable data processing pipelines** in Python and R that automated the cleaning and feature extraction of 5,000 + multi-modal neural and behavioral data, **reducing data preparation time by 40%** and enabling scalable analyses.
- Published **3 first-author papers** in leading journals (*Imaging Neuroscience, Network Neuroscience*) introducing novel modeling techniques now adopted in subsequent studies.
- **Presented data-driven findings** to interdisciplinary audiences of 200+ scientists and clinicians at 3 international conferences, directly influencing subsequent study designs.

HEAD TUTOR / ASSISTANT LECTURER | MONASH UNIVERSITY | FEBRUARY 2025 – PRESENT

- **Designed and delivered statistics and psychology curriculum** for 500+ undergraduates, utilizing data visualzsation to translate complex concepts, **resulting in an 85% personal lecturing satisfaction rate**.
- **Led a team of 12 tutors**, implementing standardized grading rubrics and feedback systems.

TEACHING FELLOW | MONASH UNIVERSITY | FEBRUARY 2024 – FEBRUARY 2025

- Collaborated with faculty to **redesign course materials and assessments**, which led to a **15% increase** in course satisfaction scores.

Education

PHD | MONASH UNIVERSITY | SUBMITTED JULY 2025

- Titled: “*High Precision Phenotypic Estimates for Brain-Behaviour Association Analyses*”

MASTERS OF PSYCHOLOGY | UNIVERSITY OF AUCKLAND | AWARDED OCTOBER 2019

- Applied signal processing techniques (Fourier transforms, wavelet analysis) to model neural oscillatory dynamics *(*awarded with First Class Honours*).*

BACHELOR OF ARTS | VICTORIA UNIVERSITY WELLINGTON | AWARDED NOVEMBER 2016